Ioanna Gkerdouki

San Diego, CA | igkerdouki.github.io | i.gkerdouki@gmail.com
linkedin.com/in/ioanna-gkerdouki | github.com/Igkerdouki | Greek (Native), English (Fluent)

EDUCATION

University of California, San Diego La Jolla, CA

Bachelor of Science, Mathematics—Computer Science Expected Mar 2027

Relevant Coursework: Machine Learning (CSE 151A), Intro to AI: Probabilistic Reasoning (CSE 150A), Advanced Data Structures in C++ (CSE 100), Abstract Algebra I & II (MATH 103A/B), Numerical Linear Algebra (MATH 170A), Combinatorics (MATH 184), Mathematical Statistics (MATH 181A), Probability (MATH 180A), Data Structures & Algorithms (Java), Systems Programming (C), Discrete Mathematics.

EXPERIENCE

San Diego Supercomputer Center (SDSC), UC San Diego San Diego, CA (Remote)

Product Owner & Software Developer Intern Jun 2026 – Present

- Originated Perceptra, an AI-powered surveillance platform that detects weapons, theft, falls, and medical emergencies in real time; defined the product vision and end-to-end system architecture.
- Built the machine learning models powering the platform's real-time detection.
- Review pull requests across the stack for architectural consistency, verifying that front-end and back-end data schemas stay aligned as the API evolves.
- Scope tasks and drive architecture decisions; designed the product landing page and visual identity.

Integrated Electronics & Biointerfaces Laboratory (IEBL), UC San Diego La Jolla, CA

Undergraduate Research Assistant Jan 2026 – Present

- Build machine learning pipelines in Python and MATLAB to decode neural signals, currently developing SVM-based color classification from optic nerve recordings.
- Preprocess multi-channel electrophysiology recordings and extract temporal and spectral features, including high-gamma band-pass filtering, envelope extraction, and RMS on intracranial EEG.

PROJECTS

WildSafe — Wildlife Collision Prevention System Winner, IEEE Quarterly Projects · Spring 2026

Raspberry Pi 4B, ML/Computer Vision, WebRTC, MQTT, Azure ML, Rust, PostgreSQL

- Pitched the project concept selected by a 5-person team from competing proposals: a distributed system that prevents wildlife–vehicle collisions on highways and flags anomalous human activity near roads for public-safety alerts.
- Architected Raspberry Pi edge nodes streaming H.264 video over WebRTC to a cloud inference endpoint, with an orchestration layer issuing MQTT commands to roadside speakers and LEDs for driver alerts.

GeoSync — Pesticide Drift Monitoring Platform In Progress

Python, FastAPI, PostgreSQL/PostGIS, WebSockets, JWT, XGBoost, React, MapLibre GL

- Building a geospatial platform over 55M+ California Pesticide Use Report records that predicts off-farm pesticide drift from a Pasquill–Gifford atmospheric stability model and live weather data.
- Built a 9-router FastAPI backend with JWT-authenticated farmer accounts and a React/MapLibre GL dashboard that broadcasts submitted spray reports over WebSockets to a live risk map.

Bloom — Portfolio Tracker & Multi-Agent Trading System In Progress

React, TypeScript, Vite, Tailwind CSS, Python, FastAPI, SQLAlchemy, Interactive Brokers API

- Built a full-stack tracker with real-time P&L, allocation and performance charts, and fundamental/technical stock analysis over a FastAPI backend with Interactive Brokers sync.
- Designed a multi-agent system (orchestrator, scanner, execution, journal) routing typed priority messages between agents with persistent memory, driven by TradingView webhook signals.

Also: UTF-8 HTTP Chat Server — concurrent HTTP server in C over raw POSIX sockets (Fall 2025).

SKILLS

Programming Languages: Python, C++, C, Java, TypeScript, MATLAB, R
Mathematics & ML: linear algebra (incl. numerical), probability & statistics, combinatorics, Bayesian networks, MLE
Signal Processing & Data: iEEG analysis, band-pass / high-gamma filtering, envelope & RMS, stimulus alignment, time-series
Web & Data: FastAPI, React, SQLAlchemy, PostgreSQL/PostGIS, REST APIs, WebSockets, JWT, XGBoost, Vite
Systems & Tools: POSIX, memory management, sockets, HTTP, concurrency, Git, Linux, Vim, GDB, IntelliJ, Eclipse